

1. Executive Summary

		Traffic Summary		Count			
Metric		Value		Total Flows Observed		13859	
Total Threats Mitigated		14		Total Packets Analyzed		5200829172	
High Priority		6		Malicious Packets		5189881328	
Low Priority		8		True Negatives Passed		9664	
Manual Releases		0					
Manual Blocks		0					
				Attack Vector		Count	
				ICMP Flood		10	
ML Metrics		Value		SYN Flood		0	
False Positives		121		UDP Flood		0	
FP Rate		1.01%		Uncertain		4	
IF Precision		97.65%					
IF Recall		98.93%		Action		Count	
IF F1-Score		98.28%		Quarantined		3	
IF Accuracy		98.97%		Time Ban		3	
				Blackhole		0	
				Released		2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	97.65%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.93%	Share of actual attacks successfully flagged
F1-Score	98.28%	Balanced measure combining Precision and Recall
Accuracy	98.97%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.07%	Actual attacks that went undetected
True Positive Rate (TPR)	98.93%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4067	FN 44
Actual: Normal	FP 98	TN 9620

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	4067	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	97.65%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.93%	Share of actual attacks successfully flagged
F1-Score	98.28%	Balanced measure combining Precision and Recall
Accuracy	98.97%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.01%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.07%	Actual attacks that went undetected
True Positive Rate (TPR)	98.93%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.99%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4067	FN 44
Actual: Normal	FP 98	TN 9620

2d. Response Latency

Metric	Value	Description
Detection Time	392.8 ms	Average time taken to detect an attack
Mitigation Response Time	15.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.74%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	1.87%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.09%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.23	3	3	2026-09-15 12:25:14	2026-09-15 12:26:36
10.0.0.24	2	2	2026-09-15 12:25:14	2026-09-15 12:26:24
10.0.0.25	2	2	2026-09-15 12:25:17	2026-09-15 12:26:28
10.0.0.1	4	1	2026-09-15 12:25:52	2026-09-15 12:26:56

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-15 12:25:14	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 12:25:14	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 12:25:17	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-15 12:25:24	10.0.0.24	DDoS	ICMP Flood	91.0%	High	Time Ban (1m)
2026-09-15 12:25:25	10.0.0.24	DDoS	ICMP Flood	91.0%	Critical	Time Ban
2026-09-15 12:25:25	10.0.0.23	DDoS	ICMP Flood	91.1%	High	Time Ban (1m)
2026-09-15 12:25:25	10.0.0.23	DDoS	ICMP Flood	91.1%	Critical	Time Ban
2026-09-15 12:25:28	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban (1m)
2026-09-15 12:25:29	10.0.0.25	DDoS	ICMP Flood	91.2%	High	Time Ban
2026-09-15 12:26:02	10.0.0.1	DDoS	Uncertain	59.8%	Low	Released
2026-09-15 12:26:24	10.0.0.24	DDoS	ICMP Flood	92.1%	High	Time Ban (2m)
2026-09-15 12:26:25	10.0.0.23	DDoS	ICMP Flood	94.1%	High	Released
2026-09-15 12:26:28	10.0.0.25	DDoS	ICMP Flood	92.1%	Critical	Time Ban (2m)
2026-09-15 12:26:36	10.0.0.23	DDoS	ICMP Flood	94.1%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.1	Uncertain	0.6097	59.8%	Low	Phase 1	1	11s	2026-09-15 12:26:56	Attack Stopped
10.0.0.1	Uncertain	0.6158	59.8%	Low	Phase 1	1	11s	2026-09-15 12:26:45	Attack Stopped
10.0.0.23	ICMP Flood	0.6356	94.1%	Critical	Phase 2	3	11s	2026-09-15 12:26:36	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6134	59.8%	Low	Phase 1	1	10s	2026-09-15 12:26:34	Attack Stopped
10.0.0.25	ICMP Flood	0.6416	92.1%	Critical	Phase 2	2	1m 11s	2026-09-15 12:26:28	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6336	94.1%	High	Phase 2	1	1m 11s	2026-09-15 12:26:25	Ban Expired
10.0.0.24	ICMP Flood	0.6419	92.1%	High	Phase 2	2	1m 10s	2026-09-15 12:26:24	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6165	59.8%	Low	Phase 1	1	10s	2026-09-15 12:26:02	Attack Stopped
10.0.0.25	ICMP Flood	0.6444	91.2%	High	Phase 2	1	11s	2026-09-15 12:25:28	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6454	91.1%	High	Phase 2	1	11s	2026-09-15 12:25:25	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.24	ICMP Flood	0.6455	91.0%	High	Phase 2	1	10s	2026-09-15 12:25:24	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE